

Midterm - CS393

1 $H \otimes I \cdot I \otimes Y \cdot CNOT(1, 0)$

$\rightarrow |0\rangle \otimes |0\rangle \quad \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \cdot |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
 $H \otimes I \quad \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \cdot |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
 $\begin{cases} Y|0\rangle = i|1\rangle \\ Y|1\rangle = -i|0\rangle \end{cases}$

$I \otimes Y \quad \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \otimes Y|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle i|1\rangle + |10\rangle -i|1\rangle)$
 $= \frac{1}{\sqrt{2}}(i|01\rangle + |11\rangle)$

$\otimes CNOT \Rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle)$

$\hookrightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \quad I \otimes Y = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{bmatrix}$

$CNOT(\alpha, \beta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad H \cdot (Y \cdot Cx)$

$Y \cdot Cx = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \end{bmatrix} \cdot H = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$

⑥ A matrix is unitary if multiplied with its adjoint = identity

$U^* = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$\boxed{2} H \otimes I \cdot I \otimes Y \cdot CX(1,0) = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \quad \boxed{3} P(|110\rangle) = 0.0$$

Probability for $|00\rangle$: 0

$$\text{Final State: } \frac{1}{\sqrt{2}}(|0\rangle|01\rangle + |0\rangle|11\rangle)$$

Probability for each is Magnitude² of their Coeff.

$$\boxed{4} H \otimes I \cdot I \otimes Y \otimes I \cdot CX(1,0) \cdot CCX(0,1:2)$$

$$\parallel \begin{matrix} |0\rangle \otimes |0\rangle \otimes |0\rangle \\ \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |00\rangle \end{matrix} = \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle)$$

$$I \otimes Y \otimes I \cdot \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}(|010\rangle + |110\rangle)$$

$$CX(0,1) \cdot \frac{1}{\sqrt{2}}(|010\rangle + |100\rangle) \rightarrow \text{Matrix in .txt}$$

$CCX(0,1:2) \rightarrow$ Has no effect $\therefore$ Same probabilities, as $\boxed{2}, \boxed{3}$ w/ $|0\rangle$ tensored (appended)
 $\hookrightarrow$ for $|000\rangle$ input.

$$\boxed{5} \quad \boxed{6} \quad \begin{matrix} \searrow \\ P|000\rangle : 0.0 \quad P|111\rangle : 0.0 \end{matrix}$$

for full prob list, see .txt.

7 $f: \{0,1\}^3 \rightarrow \{0,1\}$

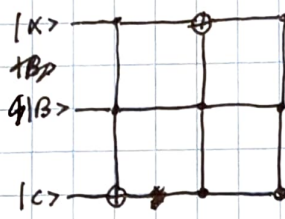
CCNOT(0,1,2)

(a)

(ii)

(a) Balanced.

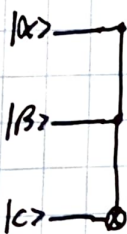
$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



b.1 - Neither

$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$
0	1	0	0
1	0	1	1
2	1	0	0
3	0	0	0
4	0	0	0
5	0	1	1
6	0	0	0
7	1	1	1



8 $S = 01 \quad \vec{x} = \vec{y} \text{ or } \vec{x} = f(x) \oplus \vec{s} \quad f(00) \mapsto 10 = f(01)$

a) How many f for $S = 01$? (Permutation)
 $f(00) = f(01) = 00$

$f(01) \mapsto 10$
 $f(10) \mapsto 00 = f(11)$
 $f(11) \mapsto 00$

4 possible outputs in $\{0, 1\}^2 \rightarrow \{0, 1\}^2$
 $00, 01, 10, 11$

$P(n, r)$ where n is number of choices, and r is how many containers.

$$P(n, r) = \frac{n!}{(n-r)!}$$

Containers: $f(00) = f(01)$ & $f(10) = f(11)$

$\therefore \# \text{ choices} = \frac{4!}{2!} = 12$

b) Space of vectors ortho to $\vec{s} \in \{0, 1\}$

~~$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0+1 = 1$~~

for $\mathbb{F}_2$, $1+1=0$ Orthogonal if dot product = 0

$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0+1=1 \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0$

$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \quad \therefore \perp: \{00, 10\}$

Also: $\{00, 01, 10, 11\} / \{00, 01\} \text{ mod } 2 = \{00, 10\}$
 Quotient Space

~~$\begin{matrix} 00 & 01 & 10 & 11 \\ 00 & 00 & 00 & 01 \\ 00 & & & \end{matrix}$~~

⑥

$$|00\rangle \otimes |00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \otimes \frac{1}{\sqrt{2}}|00\rangle$$

$$\frac{1}{2}(|00\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |01\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |10\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |11\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle))$$

$$\frac{1}{2}(|00\rangle \otimes |00\rangle + |01\rangle \otimes |00\rangle + |10\rangle \otimes |00\rangle + |11\rangle \otimes |00\rangle)$$

$$\begin{aligned} S(00) &\mapsto 10 \\ S(01) &\mapsto 10 \\ S(10) &\mapsto 00 \\ S(11) &\mapsto 00 \end{aligned}$$

$$\frac{1}{2}(|0010\rangle + |0110\rangle + |1000\rangle + |1100\rangle)$$

$H \otimes I$

$$\frac{1}{2}(H(|00\rangle)|10\rangle + H(|01\rangle)|10\rangle + H(|10\rangle)|00\rangle + H(|11\rangle)|00\rangle)$$

$$\frac{1}{2}(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle \otimes |0\rangle + |1\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle \otimes |0\rangle - |1\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle \otimes |0\rangle + |1\rangle)|00\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle \otimes |0\rangle - |1\rangle)|00\rangle)$$

$$\frac{1}{\sqrt{2}}[|0\rangle - |1\rangle \otimes |0\rangle + |1\rangle]|00\rangle + \frac{1}{\sqrt{2}}[|0\rangle - |1\rangle \otimes |0\rangle - |1\rangle]|00\rangle$$

$$\frac{1}{2}[\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)|10\rangle +$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)|00\rangle + \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)|00\rangle]$$

$$\frac{1}{2}[\frac{1}{\sqrt{2}}(|0010\rangle + |0110\rangle + |1010\rangle + |1110\rangle) \frac{1}{\sqrt{2}}(|0010\rangle - |0110\rangle + |1010\rangle - |1110\rangle) +$$

$$\frac{1}{\sqrt{2}}(|0000\rangle + |0100\rangle - |1000\rangle - |1100\rangle) + \frac{1}{\sqrt{2}}(|0000\rangle - |0100\rangle - |1000\rangle + |1100\rangle)]$$

$$\frac{1}{4}[2|0010\rangle - 2|0110\rangle + 2|1010\rangle - 2|1110\rangle + 2|0000\rangle - 2|0100\rangle - 2|1000\rangle - 2|1100\rangle]$$

$$\frac{1}{4}[|0000\rangle + |0100\rangle - |0100\rangle - |1000\rangle - |0110\rangle + |1010\rangle - |1100\rangle - |1110\rangle]$$

$$2|0010\rangle + 2|1010\rangle + 2|0000\rangle - 2|1000\rangle$$

$$\frac{1}{2}(|0010\rangle + |1010\rangle + |0000\rangle - |1000\rangle)$$

⑥ Given that:

$|0000\rangle \mapsto |0010\rangle$
 $|0100\rangle \mapsto |0110\rangle$
 $|1000\rangle \mapsto |1000\rangle$
 $|1100\rangle \mapsto |1100\rangle$

$$V_f =$$

0000 10> | 01 11> | 27 12> | 37 13> | 47 14> | 57 15> | 67 16> | 77 17> | 87 18> | 97 19> | 107 20> | 117 21> | 127 22> | 137 23> | 147 24> | 157 25> | 167 26>

0001 11> | 0010 27> | 0011 37> | 1010 47> | 1011 57> | 1100 67> | 1101 77> | 1110 87> | 1111 97> | 1200 107> | 1201 117> | 1210 127> | 1211 137> | 1300 147> | 1301 157> | 1310 167> | 1311 177> | 1400 187> | 1401 197> | 1410 207> | 1411 217> | 1500 227> | 1501 237> | 1510 247> | 1511 257> | 1600 267> | 1601 277> | 1610 287> | 1611 297> | 1700 307> | 1701 317> | 1710 327> | 1711 337> | 1800 347> | 1801 357> | 1810 367> | 1811 377> | 1900 387> | 1901 397> | 1910 407> | 1911 417> | 2000 427> | 2001 437> | 2010 447> | 2011 457> | 2100 467> | 2101 477> | 2110 487> | 2111 497> | 2200 507> | 2201 517> | 2210 527> | 2211 537> | 2300 547> | 2301 557> | 2310 567> | 2311 577> | 2400 587> | 2401 597> | 2410 607> | 2411 617> | 2500 627> | 2501 637> | 2510 647> | 2511 657> | 2600 667> | 2601 677> | 2610 687> | 2611 697> | 2700 707> | 2701 717> | 2710 727> | 2711 737> | 2800 747> | 2801 757> | 2810 767> | 2811 777> | 2900 787> | 2901 797> | 2910 807> | 2911 817> | 3000 827> | 3001 837> | 3010 847> | 3011 857> | 3100 867> | 3101 877> | 3110 887> | 3111 897> | 3200 907> | 3201 917> | 3210 927> | 3211 937> | 3300 947> | 3301 957> | 3310 967> | 3311 977> | 3400 987> | 3401 997> | 3410 1007> | 3411 1017> | 3500 1027> | 3501 1037> | 3510 1047> | 3511 1057> | 3600 1067> | 3601 1077> | 3610 1087> | 3611 1097> | 3700 1107> | 3701 1117> | 3710 1127> | 3711 1137> | 3800 1147> | 3801 1157> | 3810 1167> | 3811 1177> | 3900 1187> | 3901 1197> | 3910 1207> | 3911 1217> | 4000 1227> | 4001 1237> | 4010 1247> | 4011 1257> | 4100 1267> | 4101 1277> | 4110 1287> | 4111 1297> | 4200 1307> | 4201 1317> | 4210 1327> | 4211 1337> | 4300 1347> | 4301 1357> | 4310 1367> | 4311 1377> | 4400 1387> | 4401 1397> | 4410 1407> | 4411 1417> | 4500 1427> | 4501 1437> | 4510 1447> | 4511 1457> | 4600 1467> | 4601 1477> | 4610 1487> | 4611 1497> | 4700 1507> | 4701 1517> | 4710 1527> | 4711 1537> | 4800 1547> | 4801 1557> | 4810 1567> | 4811 1577> | 4900 1587> | 4901 1597> | 4910 1607> | 4911 1617> | 5000 1627> | 5001 1637> | 5010 1647> | 5011 1657> | 5100 1667> | 5101 1677> | 5110 1687> | 5111 1697> | 5200 1707> | 5201 1717> | 5210 1727> | 5211 1737> | 5300 1747> | 5301 1757> | 5310 1767> | 5311 1777> | 5400 1787> | 5401 1797> | 5410 1807> | 5411 1817> | 5500 1827> | 5501 1837> | 5510 1847> | 5511 1857> | 5600 1867> | 5601 1877> | 5610 1887> | 5611 1897> | 5700 1907> | 5701 1917> | 5710 1927> | 5711 1937> | 5800 1947> | 5801 1957> | 5810 1967> | 5811 1977> | 5900 1987> | 5901 1997> | 5910 2007> | 5911 2017> | 6000 2027> | 6001 2037> | 6010 2047> | 6011 2057> | 6100 2067> | 6101 2077> | 6110 2087> | 6111 2097> | 6200 2107> | 6201 2117> | 6210 2127> | 6211 2137> | 6300 2147> | 6301 2157> | 6310 2167> | 6311 2177> | 6400 2187> | 6401 2197> | 6410 2207> | 6411 2217> | 6500 2227> | 6501 2237> | 6510 2247> | 6511 2257> | 6600 2267> | 6601 2277> | 6610 2287> | 6611 2297> | 6700 2307> | 6701 2317> | 6710 2327> | 6711 2337> | 6800 2347> | 6801 2357> | 6810 2367> | 6811 2377> | 6900 2387> | 6901 2397> | 6910 2407> | 6911 2417> | 7000 2427> | 7001 2437> | 7010 2447> | 7011 2457> | 7100 2467> | 7101 2477> | 7110 2487> | 7111 2497> | 7200 2507> | 7201 2517> | 7210 2527> | 7211 2537> | 7300 2547> | 7301 2557> | 7310 2567> | 7311 2577> | 7400 2587> | 7401 2597> | 7410 2607> | 7411 2617> | 7500 2627> | 7501 2637> | 7510 2647> | 7511 2657> | 7600 2667> | 7601 2677> | 7610 2687> | 7611 2697> | 7700 2707> | 7701 2717> | 7710 2727> | 7711 2737> | 7800 2747> | 7801 2757> | 7810 2767> | 7811 2777> | 7900 2787> | 7901 2797> | 7910 2807> | 7911 2817> | 8000 2827> | 8001 2837> | 8010 2847> | 8011 2857> | 8100 2867> | 8101 2877> | 8110 2887> | 8111 2897> | 8200 2907> | 8201 2917> | 8210 2927> | 8211 2937> | 8300 2947> | 8301 2957> | 8310 2967> | 8311 2977> | 8400 2987> | 8401 2997> | 8410 3007> | 8411 3017> | 8500 3027> | 8501 3037> | 8510 3047> | 8511 3057> | 8600 3067> | 8601 3077> | 8610 3087> | 8611 3097> | 8700 3107> | 8701 311

8 1

After H:

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|00\rangle$$

$$\frac{1}{\sqrt{2}}: \frac{1}{2}(|00\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |01\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |10\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |11\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle))$$

$$\frac{1}{2}(|00\rangle|00\rangle + |00\rangle|01\rangle + |00\rangle|10\rangle + |00\rangle|11\rangle + |01\rangle|00\rangle + |01\rangle|01\rangle + |01\rangle|10\rangle + |01\rangle|11\rangle + |10\rangle|00\rangle + |10\rangle|01\rangle + |10\rangle|10\rangle + |10\rangle|11\rangle + |11\rangle|00\rangle + |11\rangle|01\rangle + |11\rangle|10\rangle + |11\rangle|11\rangle)$$

H: $\frac{1}{\sqrt{2}}$

↓ extract 1st reg

$$\frac{1}{2}(|00\rangle + |01\rangle) = |00\rangle + |10\rangle$$

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \quad \frac{1}{2}(|01\rangle + |11\rangle) = |00\rangle - |10\rangle$$

Before Measurement, 1st reg = $\frac{1}{2}(|00\rangle + |10\rangle)$

② $\frac{1}{2}: |00\rangle, \frac{1}{2}: |10\rangle$